The law of portfolio learnability

Final controlled regularization experiment - 11 October 2026

Leave the current manuscript unchanged. The controlled representation extension weakens B, and incremental net evidence for updating b is inconclusive.

Baseline replication: gross Sharpe 3.561334321625; 564 monthly returns; maximum payoff difference 2.2e-16. Annual b, lambda, complexity and initial calibration all match.

The 2 x 2 result

Primary sample: February 1979-January 2025, 552 months. Initial 1978 decision excluded. Sharpe uses concatenated monthly excess returns.

Representation	Fixed initial b	Updated b
Fixed lengthscale	A: gross 3.509 / net 2.795	B: gross 3.528 / net 2.804
Chronologically selected lengthscale	C: gross 3.467 / net 2.769	D: gross 3.495 / net 2.781

Comparison	Gross Sharpe difference [95% CI]	Net Sharpe difference [95% CI]
B-A	+0.019 [+0.003, +0.036]	+0.009 [-0.005, +0.023]
C-A	-0.042 [-0.081, -0.014]	-0.026 [-0.054, -0.005]
D-B	-0.033 [-0.065, -0.009]	-0.023 [-0.048, -0.004]
D-C	+0.028 [+0.001, +0.064]	+0.012 [-0.009, +0.038]
D-joint_cv	-0.058 [-0.411, +0.232]	+0.061 [-0.239, +0.299]

Intervals use 5,000 paired circular moving-block draws, 12 months here; 6- and 24-month sensitivity is exported. Inference conditions on the fitted paths and has no multiple-comparison correction. Updating b gives a small gross gain, but its net gain is unresolved. The selected-length rule D underperforms B net under all three block lengths.

The theorem motivates the order lambda(T) proportional to T^[-b/(b+1)]. It does not establish optimality for these finite-rank estimates, empirically calibrated constants, or data-dependent representation choices. No oracle or causal claim is made.

Financial implementation and common samples

Every policy uses the same fixed pretest exposure calibration. Signed holdings are reconstructed separately; net returns use the existing accounting_step at 25 bp per traded notional plus 30 bp/year on shorts. All subperiods carry the existing account state from 1978.

Method	Gross SR	Net SR	Mean / vol (% p.a.)	Leverage	Turnover / month	Max DD (%)
A: fixed length, fixed b	3.509	2.795	18.48 / 6.61	2.02	1.58	-14.32
B: fixed length, updated b	3.528	2.804	18.70 / 6.67	2.04	1.61	-14.50
C: selected length, fixed b	3.467	2.769	18.34 / 6.62	2.00	1.56	-14.32
D: selected length, updated b	3.495	2.781	18.55 / 6.67	2.03	1.59	-14.50
Fixed-length annual CV	3.436	2.705	21.49 / 7.94	2.38	1.99	-14.61
Joint annual CV	3.553	2.719	21.46 / 7.89	2.58	2.23	-19.23
Fixed initial lambda	3.404	2.745	17.19 / 6.26	1.86	1.39	-12.14
One-anchor full-spectrum rule	3.462	2.769	17.88 / 6.46	1.95	1.50	-13.71

Mean, volatility and drawdown above are net. Leverage is gross stock notional / NAV; turnover charges purchases and sales. The one-anchor full-spectrum rule is a separate secondary reference, never substituted for B.

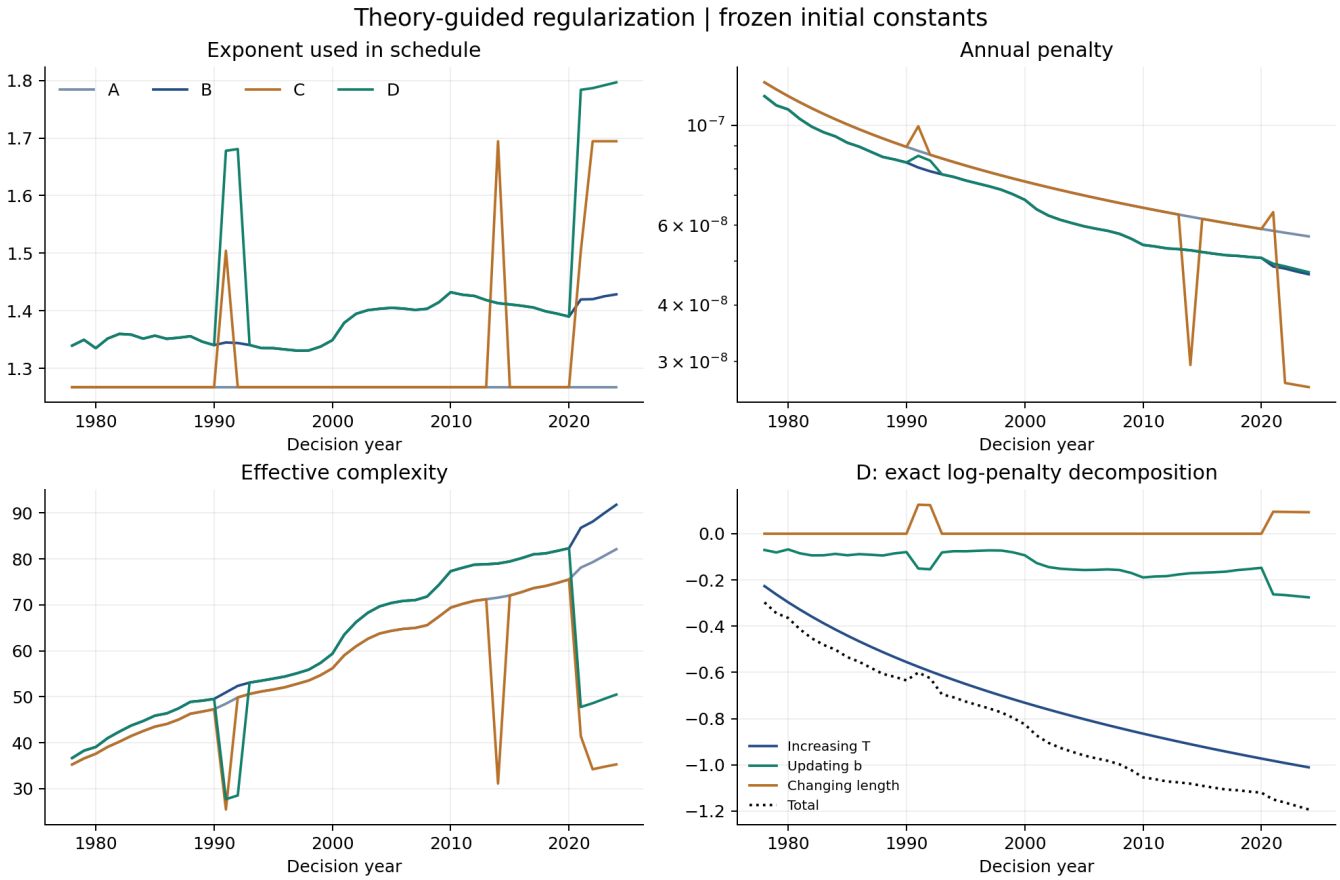
Window (February-January)	A gross/net	B gross/net	C gross/net	D gross/net	Joint CV gross/net
1978-2024	3.542/2.821	3.561/2.830	3.501/2.795	3.529/2.807	3.587/2.747
1979-2024	3.509/2.795	3.528/2.804	3.467/2.769	3.495/2.781	3.553/2.719
1984-2024	3.364/2.679	3.383/2.688	3.319/2.651	3.347/2.663	3.392/2.596
2020-2024	1.509/1.000	1.509/0.997	1.382/0.921	1.393/0.915	0.631/0.356

The window labels are decision years: e.g., 2020-2024 means February 2020-January 2025, and is descriptive only. The 1984 sensitivity removes the early overlap between initial calibration and repeated validation blocks.

Historical block	A net	B net	C net	D net	Joint CV net
1979-1989	4.664	4.666	4.664	4.666	4.637
1990-1999	4.861	4.891	4.804	4.800	4.731
2000-2009	2.260	2.277	2.260	2.277	2.516
2010-2019	1.884	1.904	1.838	1.904	2.285

These conditional historical comparisons inherit the common retrospective signal dictionary, complete-case payoff availability, current JKP snapshot and same-close execution assumptions. They do not establish future robustness.

What moves the penalty?



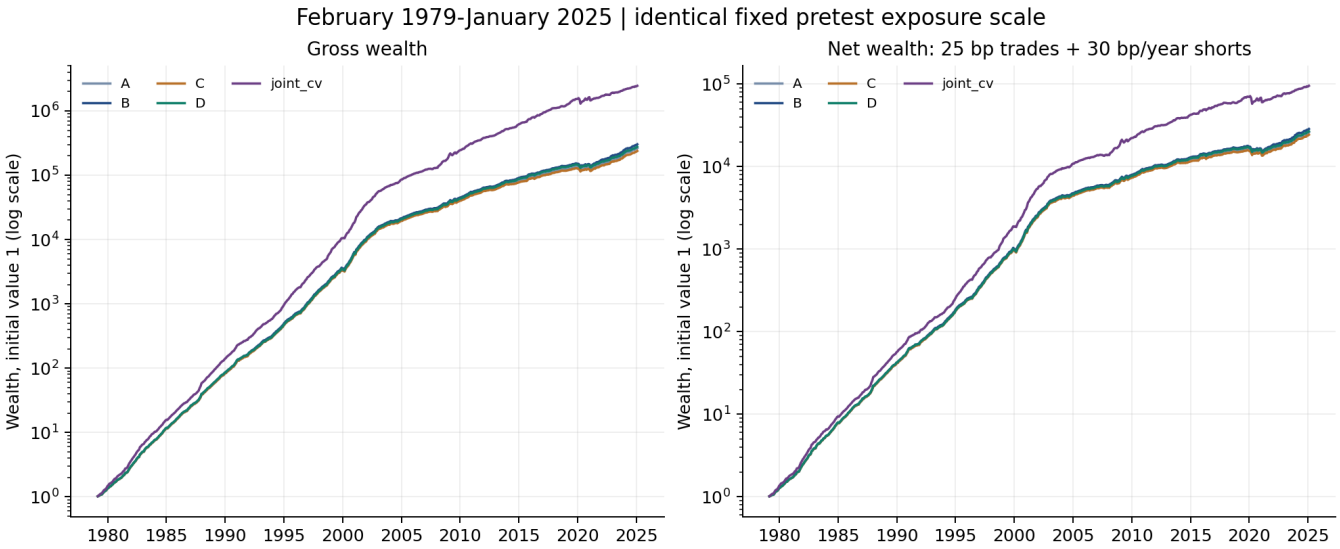
At T=732, increasing sample size accounts for 84.0% of the log decrease from the initial CV lambda; updating b accounts for 16.0%. Holding b fixed gives final lambda 5.675e-08, versus 4.68e-08 for B. These are exact lambda identities, not attributions of Sharpe.

Decomposition: log(lambda / initial baseline lambda) = sample-size term + representation term + b-update term. The representation term includes both its frozen C0 and initial exponent; the b term compares the current slope with that representation's initial slope.

Multiplier	Initial b	Initial lambda	Frozen C0	Validation loss	Admissible
0.25	0.5821	2.95e-10	1.72e-09	0.5603	No: b <= 1
0.5	0.9314	9.01e-09	9.06e-08	0.5095	No: b <= 1
1	1.2675	1.56e-07	2.27e-06	0.5986	Yes
2	1.5043	1.85e-07	3.27e-06	0.6364	Yes
4	1.6944	8.21e-08	1.67e-06	0.6327	Yes

Calibration uses 1963-1972 training (120 months) and 1973-1977 validation (60 months). C0 = initial lambda x 120^[b0/(b0+1)]; the first full-history refit uses T=180. Constants for inadmissible lengths are arithmetic extrapolations recorded for sensitivity only, never used by the primary algorithm.

Wealth, loss and admissibility



Wealth includes the frozen cash return and starts at one for the primary display. Net accounts carry the 1978 state. Terminal holdings are marked without a liquidation charge. Both A-D and the joint CV paths are exported at full monthly precision.

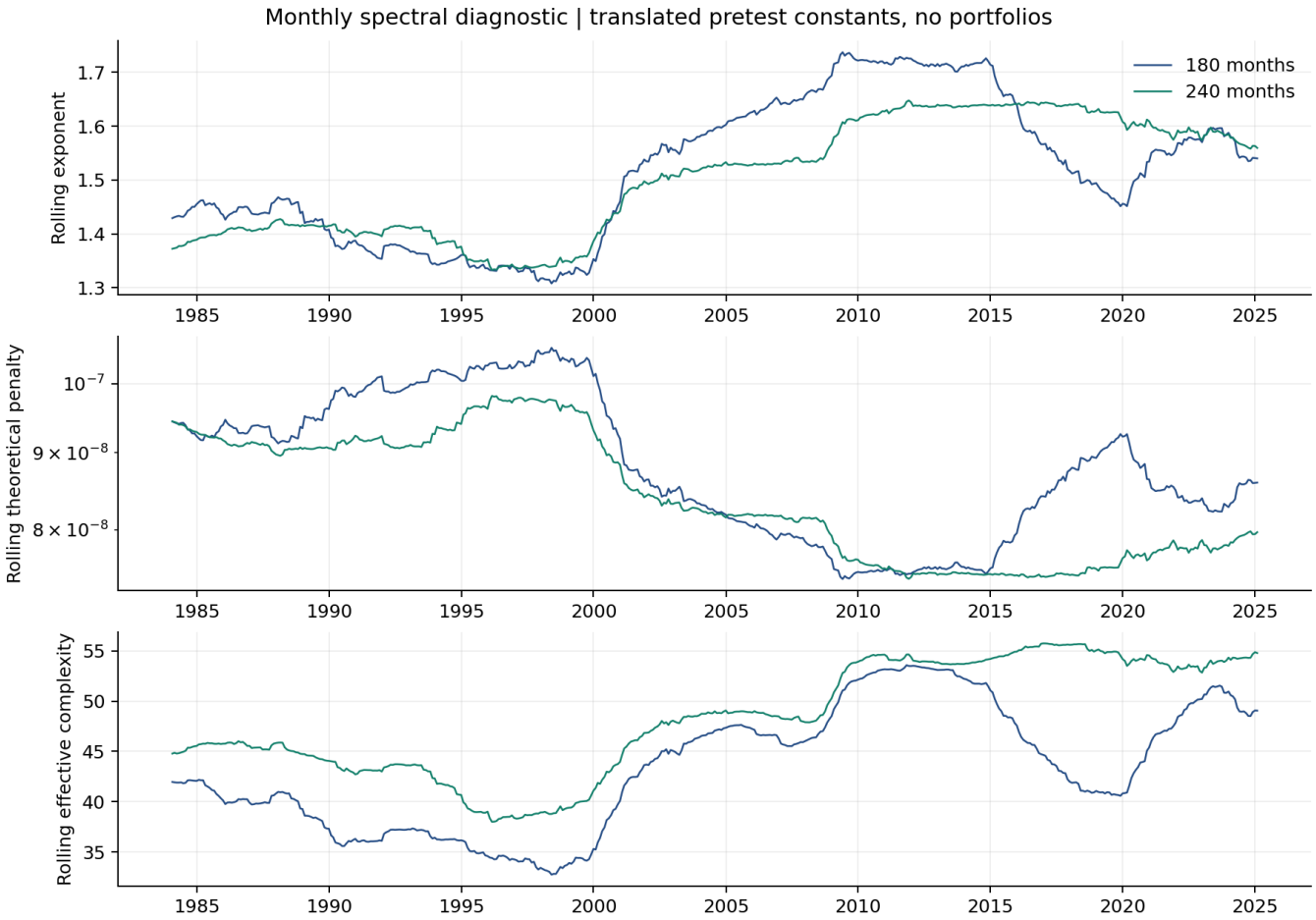
Difference	Gross response-one loss [95% CI]	Net raw-unit loss [95% CI]
B-A	-0.002 [-0.005, +0.001]	-0.001 [-0.004, +0.002]
C-A	+0.006 [+0.002, +0.012]	+0.005 [+0.001, +0.009]
D-B	+0.005 [+0.001, +0.009]	+0.004 [+0.001, +0.008]
D-C	-0.004 [-0.009, +0.001]	-0.002 [-0.006, +0.002]
D-joint_cv	-0.022 [-0.105, +0.043]	-0.034 [-0.123, +0.033]

Descriptive net interaction (D-C)-(B-A): +0.003 [-0.011, +0.023]. Lower loss is better. Net response-one losses divide monthly net excess returns by the same fixed pretest exposure scale; they are not a new selection objective.

Candidate exclusions materially affect the comparison. Two initial b estimates violate b>1. Allowing them as an explicitly outside-theorem extrapolation changes 36 of 46 primary decisions and raises D gross Sharpe from 3.495 to 3.700. This is an assumption-violation diagnostic, not a valid theorem-guided improvement; no new net claim is attached.

No lambda search occurs after initial calibration in A-D. Joint CV evaluates 600 penalty/length combinations per annual decision; C/D evaluate the three admissible lengths, each at one theory-specified penalty. Fewer validation searches do not remove the cost of eigendecomposition or portfolio refitting.

Is monthly re-estimation justified?



Trailing W	SD monthly change in b	Median |log lambda change|	Median |log complexity change|	b <= 1 frequency
180	0.0063	0.207%	0.208%	0.0%
240	0.0039	0.129%	0.140%	0.0%

Before the diagnostic begins, both window constants are translated at January 1984 close to the then-available expanding B penalty. The window-specific anchor slope determines C0(W), frozen thereafter. Four rank bands are supplied; each uses the same pretest anchoring convention.

Median monthly penalty movement is small and rank sensitivity exceeds it. Neither window meets the declared 5% monthly-change and rank-stability gate. No full monthly OOS portfolio backtest was run; the diagnostic gives no compelling reason to replace annual expanding estimation.

Holdings audit: largest scaled payoff reconciliation error 4.1e-11; accounting residual 4.4e-16; original CV net path reproduced. Numerical tests cover chronology, frozen constants, rank thresholds, primal/dual predictions, shared draws, signed holdings and figure/table reconciliation. Input hashes are saved under audit/.

Additional optimization requested during execution

Exploratory follow-up: the original five lengths plus 0.75, 1.25, 1.5 and 3 times ell0. New managed-payoff caches use the same real panels, frequency draws and phases. Each new length gets the original 200-candidate initial calibration; no annual lambda grid is added. This follow-up was specified after observing coarse-grid gross results.

Method	1979 gross/net	1984 gross/net	Last five years gross/net
C dense	3.594/2.867	3.439/2.743	1.286/0.842
D dense	3.600/2.864	3.443/2.740	1.337/0.878
C: selected length, fixed b	3.467/2.769	3.319/2.651	1.382/0.921
D: selected length, updated b	3.495/2.781	3.347/2.663	1.393/0.915
B: fixed length, updated b	3.528/2.804	3.383/2.688	1.509/0.997
Joint annual CV	3.553/2.719	3.392/2.596	0.631/0.356

Primary net comparison	Difference [12-month 95% CI]	6-month interval	24-month interval
D_dense-D	+0.083 [+0.020, +0.160]	[+0.018, +0.156]	[+0.023, +0.155]
D_dense-B	+0.060 [-0.012, +0.143]	[-0.010, +0.137]	[-0.016, +0.144]
D_dense-C_dense	-0.003 [-0.029, +0.019]	[-0.024, +0.016]	[-0.036, +0.020]
D_dense-joint_cv	+0.145 [-0.092, +0.338]	[-0.082, +0.333]	[-0.096, +0.346]

The exploratory denser search gives primary gross/net Sharpe 3.600/2.864; net D dense minus B is +0.060 [-0.012, +0.143]. It does not establish a robust incremental net advantage over B.

Added multiplier	Initial b	C0	Validation loss	Admissible
0.75	1.1412	1.14e-06	0.5628	Yes
1.25	1.3515	2.91e-06	0.6162	Yes
1.5	1.4141	3.31e-06	0.6277	Yes
3	1.6152	2.42e-06	0.6366	Yes

The expanded search evaluates nine initial grids (1,800 candidate policies), with seven initially admissible lengths. Later annual decisions select length only. Saved paired intervals condition on the resulting fitted paths and do not account for sequentially expanding the experiment after seeing earlier results.

Paper decision: retain the current manuscript. Preserve these negative, mixed and exploratory findings as a reproducible research record; there is no proposed new subsection, appendix, or oracle-optimal tuning claim.

Reproduce: empirical_final/final_theory_regularization.py; final_theory_analysis.py; final_theory_dense.py; final_theory_report.py. Public tables, candidate losses, bootstrap draws specification and source/input hashes accompany this report. Licensed stock-level holdings remain private.